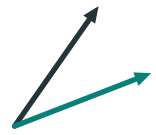


Bring them together.



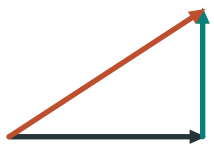
They match.



Turn one. Compare again.



Take one step, then another.

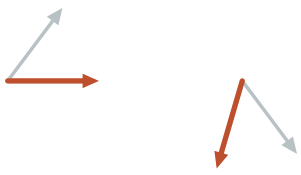


One step reaches the same end.

Follow a step back. You return.



Same lengths. Opposite green.



Turn both joined arrows the same way.



Different horizontal lengths.

Two amounts.



Move some from left to right.



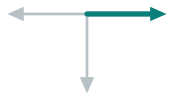
Join them: the same total.



Turn once.



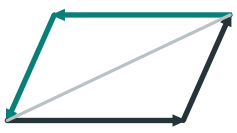
Turn again: reversed.



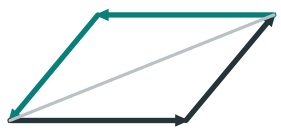
Twice more: returned.



Two triangles make a patch.



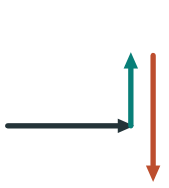
Slide the top sideways.



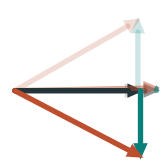
Same area. Different shape.



Now, black changes at green’s pace.



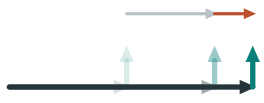
At that same instant, green changes at black’s pace, the opposite way.



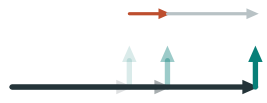
Under both rules, the joined arrow turns.



New rule: green sets the slide.



Shift, then slide.



Slide, then shift. Same end.



Black advances.



Green stays the same.



Every step keeps that amount.

Mathematics and assumptions

Companion to the nine picture strips · constructions and their added premises

Reading the pictures

Read each strip left to right; row numbers here count from the top. Black and green distinguish contributions. Orange shows a result, transferred piece or rate as specified below. Grey shows a prior position or shift; arrows in the gutters indicate reading order. Drawn geometry is a chosen representation. No background axes, physical observer at infinity, or quantum amplitudes are required to read the pictures.

1–2 · Comparison, composition and cancellation

Supply an oriented affine plane and comparison by translation. Tip-to-tail joining defines vector addition; following a step by its reverse gives zero net displacement. WDP §§1.1–1.4 requires a declared comparison structure. Its §17.7 constructs signed histories after alternative formation and additive completion are admitted. Geometric reverse steps illustrate cancellation; they do not prove every physical operation reversible. WDP §17.8 constructs $g(a,b)=L(a)^{T}L(b)$ from a supplied additive report L, with radical ker L; that pairing is observer-relative, not selected by bare difference.

3 · A report can forget what the next move needs

Inputs $\psi\pm=(3,\pm4)/5$ have equal component-length reports (3/5,4/5). Apply the SAME orthogonal matrix $U=((3,4),(-4,3))/5$: $U\psi+=(1,0)$, $U\psi-=(-7,-24)/25$. The picture uses scale 1.5 throughout. Middle-panel grey is the old joined vector; orange is the rotated vector; the last panel resolves it into components. Equal old reports give different new ones, so no report-only update works for both. A supplied square report gives (9/25,16/25) initially and first outputs 1 versus 49/625. This is not Born uniqueness, nor a failure of the l1 norm to retain signs. Sources: FQM §13.8; Proofs §§1.2–1.4.

4–5 · Balance first; a quarter-turn as an added move

Redistribute lengths (3,1)→(2,2); both total 4. More generally $z_{k+1}=z_k+d_0^* j_k$ preserves the vertex sum because each internal edge contributes opposite signs. For a closed cycle γ, $\langle A+d_0\lambda,\gamma\rangle=\langle A,\gamma\rangle$. These are balance and repair-invariance identities, not Noether's theorem (WDP §5.12). Row 5 supplies $J(x,y)=(-y,x)$: $J^2=-I$ and $J^4=I$. With an admitted rational pairing g and $g(Jx,y)=-g(x,Jy)$, WDP §17.9 derives the Q(i) action $(a+ib)x=ax+bJx$ and Hermitian form $h=g+i\,g(J\cdot,\cdot)$. J and its compatibility are inputs. Repeated direction is not elapsed time; rational turns complete to a circle, not the time line.

6 · The extra structure for classical Hamilton

Now supply a STATE plane with dimensionless conjugate coordinates q,p and area $\omega=dq\wedge dp$. The two triangles form a patch of area 5.4; the shown shears preserve it. In rows 7–9 black is q, green is p; right/up are positive. Their right angle does not derive conjugacy. Assume a time-independent continuously differentiable flow preserves EVERY small oriented area: $\partial q\, u+\partial p\, v=0$. Thus $\alpha=u\,dp-v\,dq$ is closed; on a simply connected patch $\alpha=dH$, giving $\dot q=H_p$ and $\dot p=-Hq$. H is local, unique up to a constant. The sampled shears illustrate this hypothesis; they do not establish it for arbitrary flows.

7 · A continuous rule makes the arrow turn

Choose $H=(q^2+p^2)/2$, hence $\dot q=p$ and $\dot p=-q$ simultaneously. Both rate panels show (2.1,1.2); orange rate lengths are 1.2 and 2.1 on the same scale, not successive kicks. The last panel shows exact states at t=0,0.5,1: $q(t)=q_0\cos t+p_0\sin t$, $p(t)=p_0\cos t-q_0\sin t$. It preserves H and ω. Smaller snapshots alone do not prove continuity or select a generator. This clockwise oscillator uses −J from row 5; reversing the orientation is explicit.

8–9 · A different rule, its symmetry, and the Noether charge

Switch to $H=p^2/2$: $\Phi t(q,p)=(q+tp,p)$. Translations $Ta(q,p)=(q+a,p)$ preserve ω and H, and commute with Φt. Both orderings start at (2,0.65), use a=1.5, t=1, and end at (4.15,0.65); grey means shift, orange means slide. Row 9 is the unshifted run t=0,1,2. Translation pairs with ω to give dp, so its charge is p; $\dot p=-Hq=0$. For $S=\int(p\dot q-H)dt$, independent interior variations with $\delta q=0$ at endpoints give Hamilton's equations. Constant translations preserve S and map endpoint data to translated data. Noether conservation holds on stationary trajectories; equal marks alone do not prove it.

The separate algebraic Hamiltonian route and remaining limits

WDP §19A.6 starts instead with K=Q(i), matrix multiplication, an adjoint and a K-linear derivation δ satisfying Leibniz and the adjoint law. $S=\Sigma_j\delta(E_{j1})E_{1j}$ gives $\delta=[S,\cdot]$; after skew-adjoint adjustment, $H_0=iS$ gives $\delta=-i[H_0,\cdot]$, unique modulo rational scalar identity. This constructs an algebraic generator, not a clock or flow, and is distinct from row 6's classical theorem. Neither route selects physical energy from marks. The sheets do not derive the Born rule, Everett, Bell, a unique Hilbert geometry or physical laws. Novice comprehension remains untested.